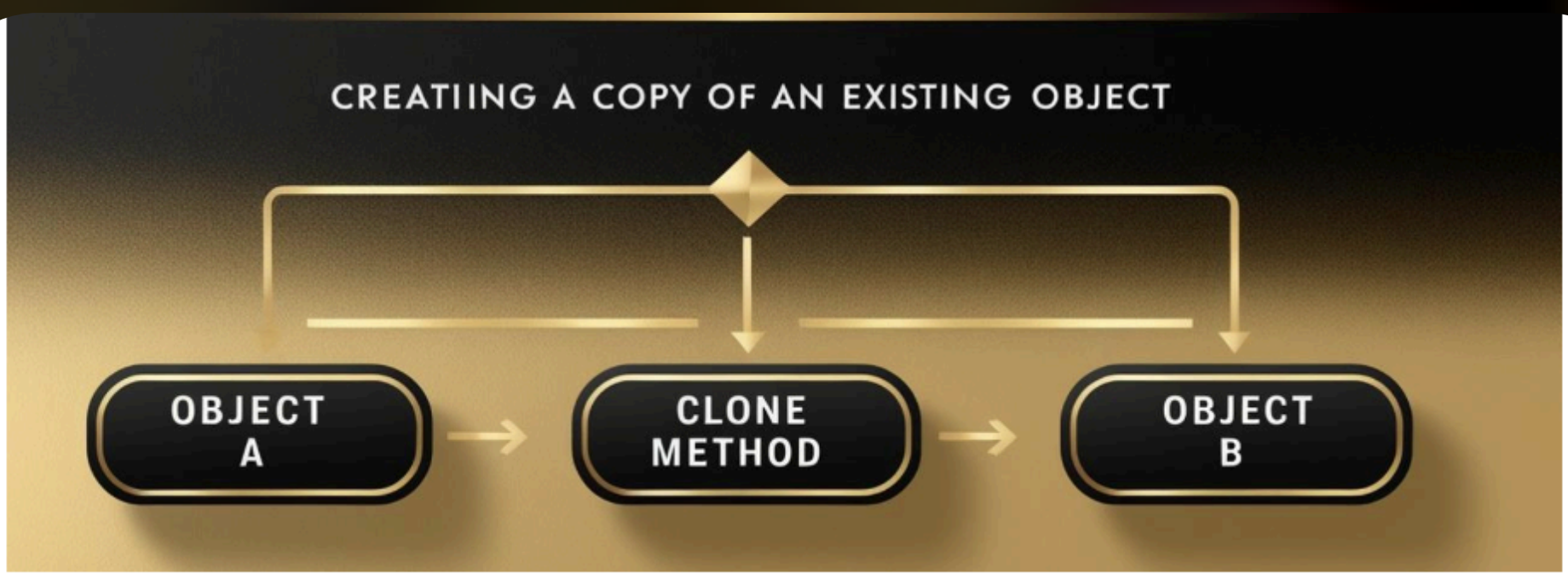
EXPLAIN CLONING IN JAVA WITH EXAMPLE



- Cloning in Java means creating a copy of an existing object. 📋✨
- It is one of the ways to create a new object in Java. 🆕🔄

For example:

- 📝 Imagine you want to create a new resume, but you don't know how to start.
- 💼 Your friend has a very good resume, so you decide to take a copy of it and edit it.
- ✅ This means you have cloned your friend's resume to create your own.

In Detail: Cloning in Java 🖥️☕

- 🆕 Cloning in Java is one of the ways to create a new object.
- 📋 When we create a new object using the cloning technique, we get an exact copy of the existing object with the same state and behaviour.
- 🛠️ To achieve this, our class must implement the Cloneable interface and override the clone() method.
- ⚠️ If not, we will encounter a CloneNotSupportedException.

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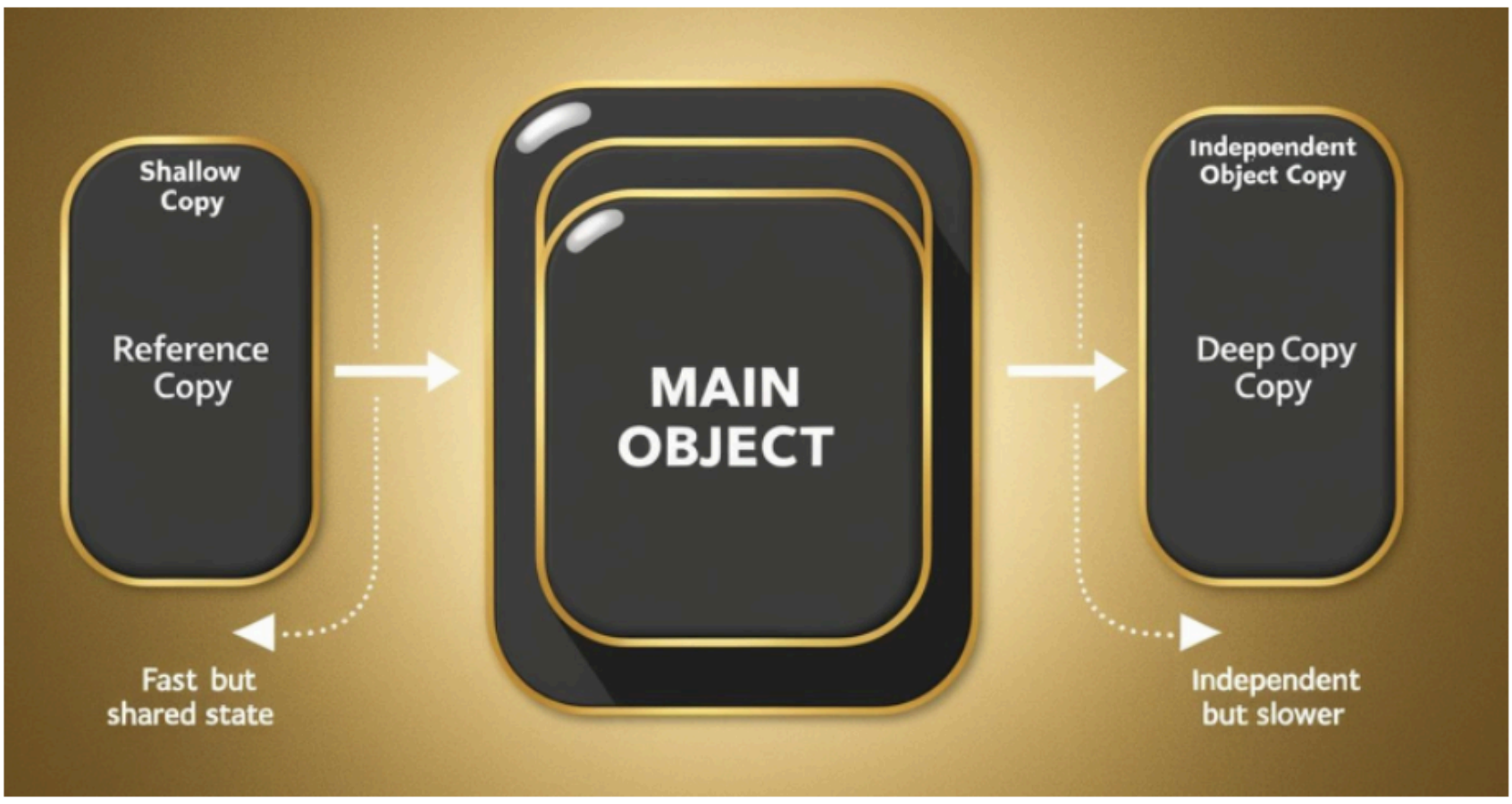
TYPES OF CLONING IN JAVA 💻☕



1. Shallow Copy 🌀

- 🔹 Creates a copy of an object using reference.
- 🔹 Copies only the reference of the object to the new object; the objects themselves are not duplicated.
- 🔹 Changes to the original object are reflected in the new object, and vice versa.
- 🔹 Performed by default when using the Object class's clone() method.

2. Deep Copy 🔄

- 🔹 Creates a copy of an object using constructor initialization.
- 🔹 New objects are created for all reference types.
- 🔹 Changes in the original object do not affect the new object, and vice versa.
- 🔹 Requires manual implementation by recursively copying all fields of the original object.

SHALLOW COPY VS. DEEP COPY

Shallow Copy vs. Deep Copy ⚖️

Shallow Copy 🌀	Deep Copy 🔄
▪ Creates a copy of an object using reference. ▪ 🚀 Fast as no new memory is allocated. ▪ 🛠️ Changes in one entity are reflected in the other entity. ▪ 📦 Default behaviour of the clone() method. ▪ 💰 Less expensive.	▪ Creates a copy of an object using constructor initialization. ▪ 🦕 Slow as new memory is allocated. ▪ 🔒 Changes in one entity do not reflect in the other entity. ▪ Requires overriding the clone() method to implement deep copy. ▪ 💸 Highly expensive.

Case Study in Cloning 📚

Are Shallow Copy and Deep Copy Related to Cloning?

- ✅ Yes, they are!
- Cloning in Java creates a new object as an exact copy of an existing object, either through shallow copy or deep copy.

How?

Shallow Copy:

- By default, the clone() method of the Object class performs a shallow copy.
- Reference fields in the cloned object point to the same objects as in the original object.

Deep Copy:

- To create a deep copy using clone(), override the method and recursively copy all fields, ensuring complete independence from the original object.

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WHEN TO USE SHALLOW COPY VS. DEEP COPY 🔍

When to Use Shallow Copy vs. Deep Copy 🔍

When to Use Shallow Copy? 🌀

- When you want a new object to share data with the original object while minimizing memory usage.
- Faster and simpler but changes in one object affect the other.
- Example: Temporary data sharing.

When to Use Deep Copy? 🔄

- When you need a new object that is completely independent of the original.
- Changes in one object do not affect the other.
- Example: When the original and copied objects require separate lifecycles.

Summary 📝

- Use deep copy when you need the cloned object to be independent of the original.
- Use shallow copy when sharing data is acceptable, and memory usage needs to be minimized.

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PURPOSE OF THIS DOCUMENT

- Use this PDF as a quick last-minute revision guide only.

- For more details, please perform R&D and gain a deeper understanding of how memory architecture works.

- Once the interview series is complete, we will cover this topic in detail.